Chapter 39 Review

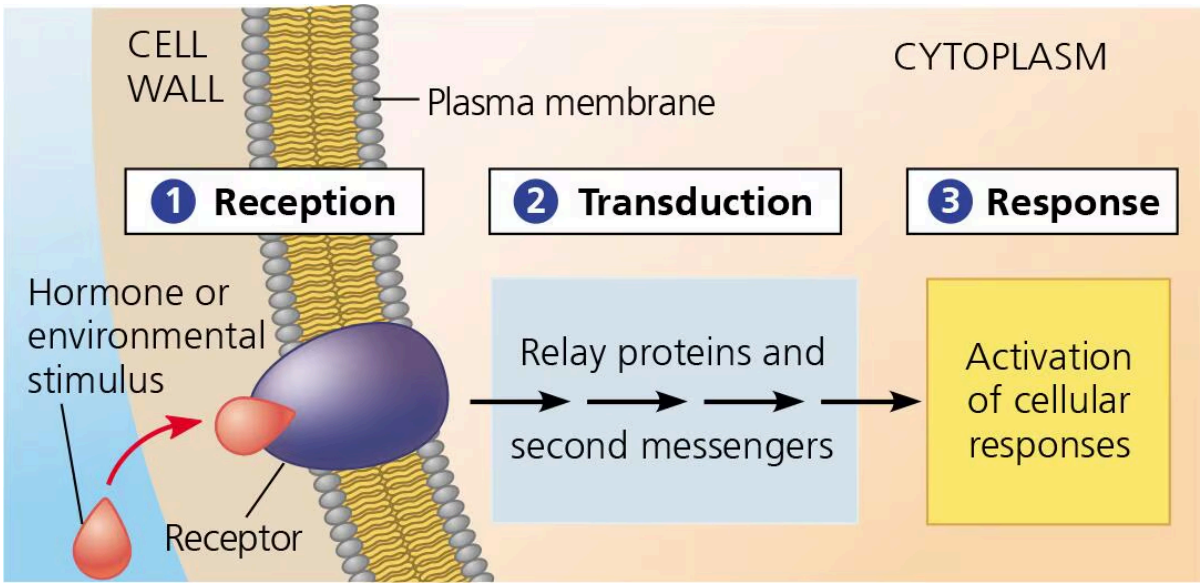
Campbell Biology, AP Edition, 12e

Chapter 39 Review

Summary of Key Concepts

Compare the experimental design of the Darwins' coleoptile experiment in 1880 with that of Boysen-Jensen in 1913. How do their hypotheses differ? **(BIG IDEA 2)**

Concept 39.1: Signal transduction pathways link signal reception to response



1. Reception: Hormone or environmental stimulus interacts with a receptor protein in the plasma membrane.
2. Transduction: Relay proteins are activated as well as second messengers that pass the signal.
3. Response: The passed signal ultimately brings about a cellular response.

What are two common ways by which signal transduction pathways enhance the activity of specific enzymes?

Concept 39.2: Plants use chemicals to communicate

- **Hormones** control plant growth and development by affecting the division, elongation, and differentiation of cells. Some also mediate the responses of plants to environmental stimuli.

Plant Hormone	Major Responses
Auxin (IAA)	Stimulates cell elongation; regulates branching and organ bending (phototropism and gravitropism)
Cytokinins	Stimulate plant cell division and differentiation; promote axillary bud growth
Gibberellins	Promote stem elongation; help seeds break dormancy and use stored reserves
Abscisic acid (ABA)	Promotes stomatal closure in response to drought; promotes seed dormancy
Ethylene	Mediates senescence, leaf abscission, fruit ripening, and obstacle avoidance by shoots (the triple response)
Brassinosteroids	Chemically similar to the sex hormones of animals; induce cell elongation and division
Jasmonates	Mediate plant defenses against insect herbivores; regulate a wide range of physiological processes
Strigolactones	Regulate apical dominance, seed germination, and mycorrhizal associations

The table is as follows:

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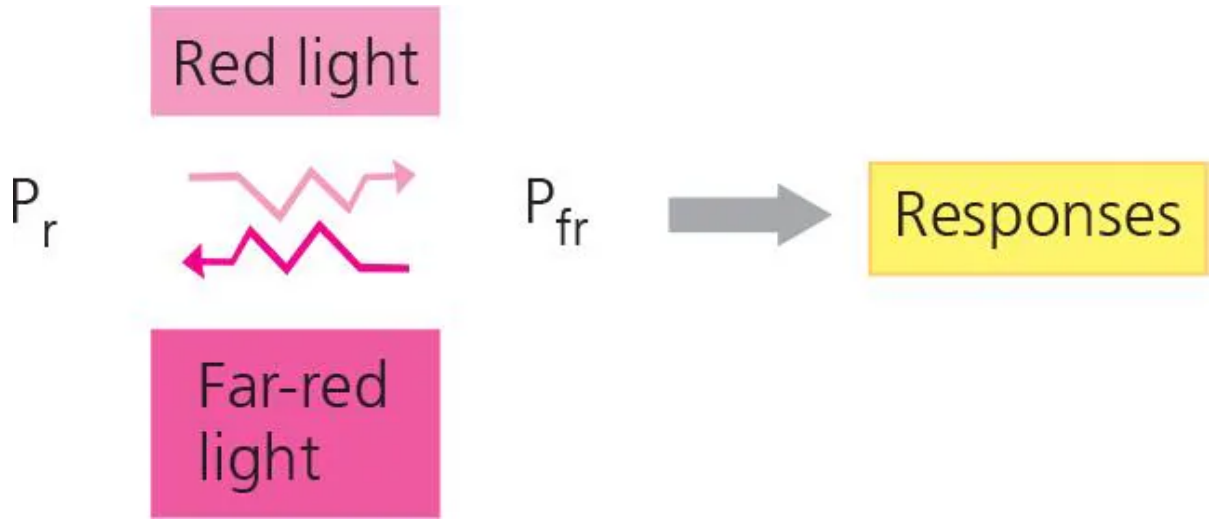
Is there any truth to the old saying "One bad apple spoils the whole bunch?" Explain.

Concept 39.3: Responses to light are critical for plant success

- ****Blue-light photoreceptors** control hypocotyl elongation, stomatal opening, and phototropism.
- ****Phytochromes** act like molecular "on-off" switches that regulate shade avoidance and germination of many seed types. Red light turns phytochrome "on," and far-red light turns it "off."



- Phytochrome conversion also provides information about the day length (photoperiod) and hence the time of year. **Photoperiodism** regulates the time of flowering in many species. **Short-day plants** require a night longer than a critical length to flower. **Long-day plants** need a night length shorter than a critical period to flower.
- Many daily rhythms in plant behavior are controlled by an internal circadian clock. Free-running **circadian rhythms** are approximately 24 hours long but are entrained to exactly 24 hours by dawn and dusk effects on phytochrome form.

Why did plant physiologists propose the existence of a mobile molecule (florigen) that triggers flowering?

Concept 39.4: Plants respond to a wide variety of stimuli other than light

- ****Gravitropism** is bending in response to gravity. Roots show positive gravitropism, and stems show negative gravitropism. **Statoliths**, starch-filled plastids, enable roots to detect gravity.
- ****Thigmotropism** is a growth response to touch. Rapid leaf movements involve transmission of electrical impulses.
- Plants are sensitive to environmental stresses, including drought, flooding, high salinity, and extremes of temperature.

Environmental Stress	Major Response
Drought	ABA production, reducing water loss by closing stomata
Flooding	Formation of air tubes that help roots survive oxygen deprivation
Salt	Avoiding osmotic water loss by producing solutes tolerated at high concentrations
Heat	Synthesis of heat-shock proteins, which reduce protein denaturation at high temperatures
Cold	Adjusting membrane fluidity; avoiding osmotic water loss; producing antifreeze proteins

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Plants that have acclimated to drought stress are often more resistant to freezing stress as well. Suggest a reason why.

Concept 39.5: Plants respond to attacks by pathogens and herbivores

- The **hypersensitive response** seals off an infection and destroys both pathogen and host cells in the region. **Systemic acquired resistance** is a generalized defense response in organs distant from the infection site.
- In addition to physical defenses such as thorns and trichomes, plants produce distasteful or toxic chemicals, as well as attractants that recruit animals that destroy herbivores.

How can insects make plants more susceptible to pathogens?